

SHOAL LAKE, MB, Canada

WMO: 711500

Lat: 50.4550N

Lon: 100.6036W

Elev: 561

StdP: 94.76

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-33.5	-30.8	-37.0	0.1	-33.6	-34.0	0.2	-30.8	13.4	-11.6	12.0	-13.0	2.8	300	0.724

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.2	28.4	19.9	26.7	19.0	25.1	17.9	22.0	26.2	20.6	24.6	19.2	23.3	5.1	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
20.7	16.4	24.3	19.1	14.9	22.8	17.8	13.7	21.6	67.5	26.2	62.1	24.4	57.4	23.3	28.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.6	10.3	9.1	-37.5	31.7	2.8	1.7	-39.5	32.9	-41.2	33.9	-42.8	34.9	-44.8	36.2		
			-37.6	24.2	2.8	2.0	-39.6	25.6	-41.3	26.8	-42.8	27.9	-44.9	29.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	1.6	-16.8	-14.9	-7.6	2.3	9.8	15.3	17.8	16.7	11.8	3.8	-5.8	-13.7	(6)
(7)		DBStd	13.76	8.31	7.79	7.84	6.39	4.73	3.59	2.99	3.45	4.41	4.94	6.68	7.73	(7)
(8)		HDD10.0	3818	830	697	546	237	62	3	0	1	31	200	475	735	(8)
(9)		HDD18.3	6161	1088	931	804	480	267	102	46	73	201	451	725	993	(9)
(10)		CDD10.0	768	0	0	0	7	56	163	243	208	83	8	0	0	(10)
(11)		CDD18.3	69	0	0	0	0	2	12	31	22	3	0	0	0	(11)
(12)		CDH23.3	786	0	0	0	1	43	124	280	269	66	3	0	0	(12)
(13)		CDH26.7	151	0	0	0	0	8	21	49	64	9	0	0	0	(13)
(14)	Wind	WSAvg	4.3	4.2	4.2	4.5	4.8	5.1	4.3	3.7	3.7	4.2	4.6	4.5	4.1	(14)
(15)	Precipitation	PrecAvg	506	26	17	28	26	64	94	75	59	41	29	23	27	(15)
(16)		PrecMax	706	52	54	65	54	172	228	139	128	58	61	58	54	(16)
(17)		PrecMin	347	2	2	2	8	8	36	8	21	17	10	5	9	(17)
(18)		PrecStd	105	14	13	17	13	40	50	32	31	12	17	15	13	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.2	3.1	12.1	21.9	27.5	29.0	30.1	31.1	28.0	22.1	11.4	3.7	(19)
(20)			MCWB	0.8	1.8	7.5	12.1	15.3	19.8	22.6	20.7	16.9	13.1	7.2	1.7	(20)
(21)		2%	DB	-0.4	0.7	7.2	18.0	24.1	26.5	28.0	28.5	25.3	17.5	7.4	0.9	(21)
(22)			MCWB	-1.2	0.0	4.3	9.7	13.3	17.9	21.2	19.8	16.8	10.9	4.4	-0.1	(22)
(23)		5%	DB	-2.4	-1.2	4.2	15.2	21.4	24.5	26.3	26.3	22.8	14.2	4.7	-1.5	(23)
(24)			MCWB	-3.0	-1.8	2.4	8.4	12.2	16.7	20.2	18.7	15.6	9.1	2.6	-2.2	(24)
(25)		10%	DB	-4.6	-3.5	2.1	12.2	18.8	22.6	24.7	24.3	20.0	11.4	2.5	-3.3	(25)
(26)			MCWB	-5.0	-4.0	1.0	7.0	11.1	16.0	19.0	17.6	13.9	7.9	1.2	-3.8	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.8	2.0	7.6	13.0	17.5	22.9	24.9	23.3	20.3	13.9	7.8	2.0	(27)
(28)			MCDWB	1.8	2.9	11.4	19.4	23.3	26.4	27.5	28.8	24.2	20.0	11.3	3.5	(28)
(29)		2%	WB	-1.2	0.1	4.5	10.8	15.2	19.8	22.6	21.2	18.0	11.7	4.9	-0.1	(29)
(30)			MCDWB	-0.5	0.7	7.0	16.7	21.0	23.5	26.1	26.0	22.8	16.5	6.9	0.9	(30)
(31)		5%	WB	-3.0	-1.7	2.6	9.0	13.5	18.1	21.2	19.8	16.4	9.8	2.8	-2.1	(31)
(32)			MCDWB	-2.5	-1.2	4.2	14.4	19.2	22.2	25.1	24.1	21.3	13.1	4.4	-1.4	(32)
(33)		10%	WB	-5.0	-3.9	1.1	7.3	12.0	16.9	19.8	18.6	14.7	8.2	1.3	-3.7	(33)
(34)			MCDWB	-4.7	-3.4	2.1	11.8	17.5	21.2	23.6	22.9	18.9	11.5	2.5	-3.2	(34)
(35)	Mean Daily Temperature Range	MDBR	9.5	10.1	10.3	11.0	12.0	10.7	11.2	12.5	12.0	9.9	8.5	8.6	(35)	
(36)		5% DB	MCDBR	10.1	9.4	10.7	15.3	16.2	13.6	13.4	15.4	16.3	15.2	11.3	9.8	(36)
(37)			MCWBR	9.6	8.4	8.0	8.5	7.5	6.7	7.5	7.9	8.6	9.0	8.4	9.0	(37)
(38)		5% WB	MCDWB	10.2	9.0	10.2	14.4	13.4	11.5	12.2	12.9	14.2	13.8	10.8	10.2	(38)
(39)			MCWBR	9.6	8.2	7.8	8.4	7.2	6.7	7.7	7.6	8.3	8.6	8.3	9.3	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.257	0.255	0.316	0.332	0.342	0.355	0.374	0.379	0.346	0.281	0.252	0.226	(40)	
(41)		taud	2.252	2.294	2.228	2.382	2.415	2.410	2.371	2.345	2.414	2.573	2.483	2.374	(41)	
(42)		Ebn at Noon	783	896	883	912	914	901	877	853	846	867	811	792	(42)	
(43)		Edh at Noon	68	86	112	108	110	112	114	112	94	66	55	52	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.36	2.53	3.92	5.08	5.75	5.94	6.33	5.13	3.80	2.24	1.22	0.96	(44)	
(45)		RadStd	0.10	0.19	0.29	0.34	0.47	0.45	0.34	0.44	0.31	0.29	0.17	0.09	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (16 neighbors)	N/A	N/A	N/A	N/A	N/A	-0.50	N/A	N/A	N/A	N/A

Nomenclature: See separate page